

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q19-1233

North Campus Unm 100 kV
August 22, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1233. The Interconnection Customer proposes a 68.9 MW solar photovoltaic facility interconnecting at the North Campus Unm 100 kV substation in the Ohio area. The request was received on June 23, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 292 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 17 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$78,687,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sith Mtn – Havilah 100 KV	Beaver – Eufaula 500 KV	195.1	100	57.4
Second Fork – Pickford 100 KV	Adams – Onancock Diesel Generation Facility 230 KV	156.1	151	5.2
Smith Mtn – Green Hill 100 KV	Beaver – Eufaula 500 KV Line	152.5	116	57.4
W Sta J Substation – Havilah 100 KV	Oberlin Oh – Emory Road 100 KV line	152.2	111	54.7
North Campus Unm – West Station Nj 100 KV Line	Emory Road – Oberlin Oh 100 KV	150.6	120	54.7
Bayside – Emory Rd 100 KV Line	West Sta Nj – N Campus Unm 100 KV Line	142.9	120	51.6
Oberlin Oh – Emory Rd 100 KV	West Station Nj – N Campus Unm 100 KV	142.7	124	51.5
Ripp – Idylwild 100 KV	Smith Mtn – Havilah 100 KV	142.7	116	8.2
Ryan – West Medina 100 KV Line	Beaver – Eufaula 500 KV	141.3	139	17.3
Velco Highgate – Oberlin Oh 100 KV	West Station Nj – North Campus Unm 100 KV	141.0	118	51.5
Havilah – Velco Highgate Substation 100 KV Line	North Campus Unm – W Sta Nj 100 KV	140.4	114	51.5
Bronx – Pickford 100 KV L	Pickford – Second Fork Substation 100 KV	136.2	136	5.5
Axtel – Ripp 100 KV line	Smith Mountain – Havilah 100 KV	134.7	142	8.2
Whiskeytown – E Franklin 100 KV	Smith Mountain – Havilah 100 KV	130.0	174	8.2

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Heritage – Green Hill 100 KV line	Beaver – Eufaula 500 KV	127.5	126	57.4
W Medina – Redhills Generating Facility 100 KV	Beaver – Eufaula 500 KV line	121.4	128	17.3
Rock Rapids – Idylwild 100 KV	Smith Mtn – Havilah Substation 100 KV	107.1	129	8.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Smit Mtn Substation – Havilah 100 KV	Rebuild line with higher-capacity conductor	\$3,666,000	100% (\$3,666,000)
Second Fork – Pickford 100 KV	Rebuild line with higher-capacity conductor	\$2,861,000	100% (\$2,861,000)
Smith Mtn – Green Hill 100 KV line	Rebuild line with higher-capacity conductor	\$5,354,000	100% (\$5,354,000)
W Sta J Substation – Havilah 100 KV	Rebuild line with higher-capacity conductor	\$3,150,000	100% (\$3,150,000)
W Station Nj – N Campus Unm 100 KV	Rebuild line with higher-capacity conductor	\$10,476,000	100% (\$10,476,000)
Emory Rd Substation – Baysie 100 KV	Rebuild line with higher-capacity conductor	\$1,120,000	100% (\$1,120,000)
Oberlin Oh – Emory Rd 100 KV	Rebuild line with higher-capacity conductor	\$900,000	100% (\$900,000)
Idylwild – Ripp 100 KV line	Rebuild line with higher-capacity conductor	\$4,358,000	100% (\$4,358,000)
W Medina – Ryan 100 KV	Rebuild line with higher-capacity conductor	\$2,722,000	100% (\$2,722,000)
Velco Highgate – Oberlin Oh 100 KV	Rebuild line with higher-capacity conductor	\$5,074,000	100% (\$5,074,000)
Havilah – Velco Highgate 100 KV Line	Rebuild line with higher-capacity conductor	\$5,215,000	100% (\$5,215,000)
Bronx – Pickford 100 KV line	Rebuild line with higher-capacity conductor	\$15,271,000	100% (\$15,271,000)
Axtel – Ripp 100 KV	Rebuild line with higher-capacity conductor	\$2,458,000	100% (\$2,458,000)
Whiskeytown – E Franklin 100 KV	Rebuild line with higher-capacity conductor	\$5,668,000	100% (\$5,668,000)
Heritage – Green Hill 100 KV line	Rebuild line with higher-capacity conductor	\$5,212,000	100% (\$5,212,000)
W Medina – Redhills Generating Facility 100 KV	Rebuild line with higher-capacity conductor	\$4,590,000	100% (\$4,590,000)
Rock Rapids – Idylwild 100 KV	Reconductor line	\$592,000	100% (\$592,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$78,687,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.